
Depth-Conditioned Pseudo-Labels for Scene-Centric Unsupervised Panoptic Segmentation

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Abstract

Existing unsupervised scene-centric panoptic methods rely on calibrated stereo and unsupervised optical flow, foreclosing the long tail of monocular footage from phones, dashcams, and fixed cameras. We close this gap in two stages. First, we show that frozen monocular priors alone are sufficient to train a panoptic network to 32.76 PQ on Cityscapes val under the adopted CUPS-style trainer [17]. Second, we contribute two compact mechanisms that compose these priors through cross-modal agreement — DCFA, a $\sim 40\text{K}$ -parameter residual adapter that conditions a frozen unsupervised semantic code on monocular depth; and SIMCF, a learning-free filter that enforces semantic, geometric, and appearance agreement on candidate panoptic labels — improving this controlled monocular baseline by +3.07 PQ. The resulting 35.83 PQ also surpasses the published stereo+flow CUPS configuration [17] (27.80 PQ) under the reported CUPS evaluation protocol [17], while using only single monocular target images at pseudo-label time.

1 Introduction

Panoptic segmentation assigns a category to every pixel and a distinct identity to every countable object. The task underpins autonomous driving, robotic manipulation, and dense scene parsing, and its supervised counterpart sits at the top of every benchmark that supplies dense human masks. The obstacle is annotation cost: Cityscapes-style annotation requires a pixel-level mask for every visible instance, and the cost compounds with each new dataset and domain. The unsupervised setting asks whether a model can recover panoptic structure from raw pixels alone.

Recent progress has come from three lineages. Self-supervised semantic discovery on top of frozen DINO/DINOv2 features [7, 8] now yields region-level partitions coherent enough to serve as semantic pseudo-labels [22, 10]. Cut-and-learn instance discovery [32, 16, 15] extracts foreground masks from the same SSL features and bootstraps detectors from them. CutS3D [21] moves this instance line toward scene-centric data by lifting depth to a 3-D point cloud and applying LocalCut to disambiguate adjacent instances. CUPS [17] combines semantic pseudo-labels, stereo-derived depth, and unsupervised scene flow, then trains a Cascade Mask R-CNN with DropLoss bootstrapping and EMA self-training.

CUPS [17] obtains this structure from calibrated stereo at pseudo-label time and unsupervised optical flow over short video clips. These inputs are not available for many scene-centric datasets, where images often arrive as single monocular frames from a phone, dashcam, or fixed camera. A method that depends on multi-view input therefore narrows the range of imagery where unsupervised panoptic segmentation can be applied. We ask whether this extra input is necessary, or whether monocular images already contain enough structure for pseudo-label generation.

We show that frozen monocular priors can replace stereo and motion at pseudo-label time when they are combined through cross-modal agreement. Semantic codes provide class-consistent regions but may merge adjacent same-class objects or split a surface under illumination change. Monocular depth provides physical separations but may over-fragment objects along material seams. Dense

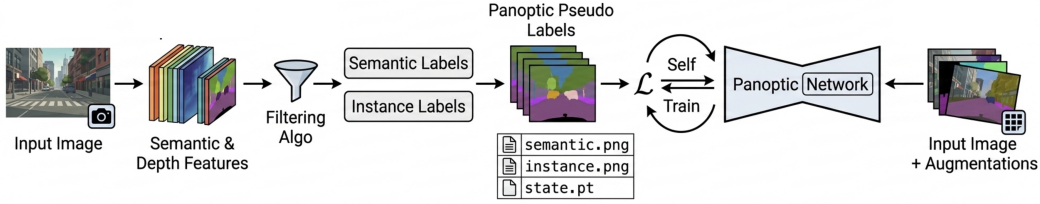


Figure 1: Pipeline overview. Stage 1 constructs panoptic pseudo-labels from a single RGB image using frozen semantic features, monocular depth, DCFA, and SIMCF. Stage 2 adopts CUPS-style [17] panoptic bootstrapping and panoptic network training on the filtered labels.

feature similarity can merge depth fragments when their appearance is compatible. Our pipeline uses these cues conservatively: depth proposes candidate boundaries, while semantic identity and feature similarity decide which boundaries should survive.

This design improves pseudo-label panoptic quality from 24.54 to 25.85 on Cityscapes before panoptic bootstrapping. After CUPS-style [17] panoptic bootstrapping and panoptic network training, the final model reaches 35.83 PQ, 62.79 SQ, and 43.78 RQ on Cityscapes val. The corresponding monocular baseline without DCFA/SIMCF reaches 32.76 PQ under the same training procedure, isolating a +3.07 PQ gain from the proposed pseudo-label generator. The published CUPS configuration [17] reaches 27.80 PQ with stereo + motion cues at pseudo-label time.

We make three contributions:

- Stereo-and-motion-free pseudo-label generation for scene-centric panoptic segmentation. Monocular semantic appearance and monocular depth, composed by an agreement filter, supply enough structure for scene-centric unsupervised panoptic-label generation, removing the stereo + motion requirement in CUPS [17] at pseudo-label time.
- Two compact mechanisms for cross-modal agreement. A depth-conditioned residual adapter (DCFA) adapts a frozen 90-dimensional semantic code toward depth-consistent regions and lifts semantic mIoU from 52.69 to 55.29; Semantic Instance Mutual Consistency Filtering (SIMCF) accepts, revises, or rejects candidate labels through semantic, geometric, and appearance agreement and contributes the bulk of the +1.31 PQ pseudo-label gain.
- 35.83 PQ on Cityscapes after panoptic network training. Using the adopted CUPS [17] panoptic bootstrapping and network-training procedures, the final network reaches 35.83 PQ on Cityscapes val, a +3.07 PQ gain over the corresponding monocular baseline.

Figure 1 gives the full pipeline view. Section 2 surveys the unsupervised semantic, instance, and panoptic literatures. Section 3 develops the pipeline. Section 4 reports Cityscapes results, ablations, and transfer experiments. Section 5 discusses the results. Section 6 summarizes the main limitations.

2 Related Work

Unsupervised Semantic Segmentation. Early unsupervised semantic segmentation methods optimized clustering objectives over learned image representations: IIC [23] maximized mutual information between augmented views, and PiCIE [24] enforced photometric and geometric invariance to recover pixel-level groupings. The arrival of self-supervised vision transformers [7, 8] reshaped the field, with dense ViT features clustering cleanly enough that *distillation* methods became dominant. STEGO [22] distilled DINO correspondences into compact dense embeddings; HP [25] improved this contrastive pool by discovering hidden positives; CAUSE [10] structured the latent space through a concept clusterbook, producing a concept-coherent semantic code. Recent variants refine the clustering objective on top of these SSL features but do not change what is being clustered: spectral aggregation in EAGLE [26], proxy-anchor mining in PPAP [27], diffusion-feature partitioning in DiffCut [28], and recursive hierarchy clustering [29].

Monocular depth has been considered as an auxiliary signal throughout this lineage. DepthG [20] adds a depth-feature correlation loss together with depth-guided positive/negative sampling on top of a STEGO-style backbone, while CUPS [17] uses depth as a contrastive auxiliary in its DINO-distillation semantic head. Hoyer *et al.* [30] use self-supervised depth as a multi-task pre-training source for semi-supervised semantic segmentation, transferring depth-trained backbone weights to a labeled-data segmentation task. DCFA differs in two ways: it operates fully unsupervised, and it conditions a frozen unsupervised concept-clusterbook code via a parameter-efficient residual rather than transferring depth-pretrained backbone weights. In a separate, fully supervised line, RGB-D fusion methods inject depth into attention but require dense labels (DFormerv2 [31]). DCFA differs from these prior unsupervised depth-guided methods along two axes. First, the conditioning site: DepthG [20] injects depth via a backbone-level correlation loss that updates the backbone end-to-end, whereas DCFA freezes the backbone and applies a feature-level residual adapter to the frozen clusterbook code (CAUSE-TR [10]), with $\sim 40\text{K}$ trainable parameters. Second, what is being conditioned: DepthG [20] re-shapes a STEGO-style code during training, whereas DCFA conditions an already-trained, frozen unsupervised code without altering its underlying manifold. The two design points are complementary; DCFA is, to our knowledge, the first feature-level residual adapter applied to a frozen unsupervised concept-clusterbook code for monocular depth conditioning.

Unsupervised Instance Segmentation. The dominant line of unsupervised instance discovery applies graph-cut-style partitioning to self-supervised features. LOST [32] localized a single salient object per image; TokenCut [16] generalized this through normalized cuts on DINO patch tokens; MaskCut [15] iterated TokenCut to extract multiple masks per image; CutLER [15] turned MaskCut pseudo-masks into supervision for a class-agnostic Cascade Mask R-CNN, establishing the cut-and-learn template. MaskDistill [33] and FreeSOLO [34] explored alternative pseudo-mask formulations within the same paradigm.

Recent extensions diversify the cut-and-learn template. CuVLER [35] strengthens MaskCut with multi-SSL feature voting and soft-target distillation; COLER [36] cuts in a single pass without k-means; ProMerge [37] introduces prompt-and-merge grouping with background-based pruning; unMORE [38] learns existence/center/boundary fields; UnSAM and UnSAMv2 [39, 40] add granularity-controlled segmentation on SA-1B-scale data. Slot-attention models such as DINOSAUR [41] and its motion-refined extension MR-DINOSAUR [42] are adjacent alternatives that largely target object-centric or class-agnostic instance discovery rather than the instance branch of scene-centric panoptic pseudo-labeling.

A separate subline injects geometric information explicitly, and CutS3D [21] is the closest comparison to our instance pipeline. CutS3D lifts each image to a 3-D point cloud through monocular depth, constructs a 3-D k-NN graph, and applies LocalCut to disambiguate adjacent instances that 2-D NCut merges. We share CutS3D’s monocular premise but stay entirely in 2-D: depth gradients provide a 2-D proxy for the separation cue CutS3D obtains after unprojecting depth to 3-D, namely local discontinuities between physically distinct surfaces. After Sobel filtering and connected-component analysis within thing-eligible semantic regions, this 2-D treatment supplies the instance side of our pseudo-labels without lifting to a point cloud or constructing a 3-D graph.

Unsupervised Panoptic Segmentation. Unsupervised panoptic segmentation is a young task. The most directly comparable Cityscapes results to date come from three works. U2Seg [18] is the first unified framework, composing STEGO and MaskCut on object-centric data and transferring to scene-centric Cityscapes; it inherits the dominant-foreground bias of MaskCut and reaches roughly 18 PQ. CUPS [17] is the current scene-centric state of the art and constructs pseudo-labels from pseudo-label-time stereo and motion cues — stereo video, unsupervised optical flow [43], and SF2SE3 scene-flow segmentation [44] — then trains a Cascade Mask R-CNN with DropLoss bootstrapping and three rounds of EMA self-training. S2-UniSeg [19] is the closest concurrent monocular alternative on Cityscapes panoptic. It reaches 25.4 PQ after segmentation-oriented pretraining on SA-1B ($\sim 11\text{M}$ images) followed by adaptation to Cityscapes train (2,975 images); we reach 35.83 PQ using Cityscapes train alone, surpassing S2-UniSeg by +10.43 PQ with a $\sim 3,700\times$ smaller pretraining corpus. Our setting differs from these concurrent monocular alternatives by generating pseudo-labels from target-dataset monocular frames alone, without stereo/video capture, explicit 3-D reconstruction, or SA-1B-scale segmentation pretraining. We inherit CUPS-style [17] panoptic bootstrapping and panoptic network training, and replace only the pseudo-label generator.

3 Method

3.1 Problem Setup and Pipeline Overview

We address scene-centric unsupervised panoptic segmentation in the monocular variant of the CUPS setting [17], where a model must produce dense panoptic predictions without human annotations. Concretely, training is given an unlabeled image set $\mathcal{D}_{\text{train}} = \{x_i\}_{i=1}^N$, in which each frame $x_i \in \mathbb{R}^{H \times W \times 3}$ is a single RGB image drawn from the target dataset, here Cityscapes [1]. The setting provides no semantic, instance, or panoptic annotations, no ground-truth or sensor-provided depth maps, no calibrated stereo views, and no temporal neighbours. We use only unlabeled target images for pseudo-label construction and panoptic network training, while relying on frozen off-the-shelf visual and depth priors. Formally, the two stages can be written as

$$G(x) = (\hat{S}_x, \hat{I}_x), \quad F(x) = (S_x, I_x), \quad (1)$$

where G is the monocular pseudo-label generator used during training and F is the panoptic network evaluated at test time. The maps $\hat{S}_x$ and S_x assign semantic labels to pixels, while $\hat{I}_x$ and I_x assign instance identities to pixels belonging to countable object classes.

Figure 1 summarizes the resulting two-stage pipeline. Working strictly within this monocular regime, we adopt the same two-stage decomposition that organizes CUPS [17]. The first stage fits any pseudo-labeling components using only unlabeled target images, then applies the resulting generator independently to each frame. The second stage follows the panoptic bootstrapping and panoptic network training procedures of CUPS [17] on these pseudo-labels. Holding this training stage fixed lets us attribute changes in final network performance to the pseudo-label source, which is also the only place where stereo and motion cues entered the original CUPS recipe [17].

The semantic branch reuses two off-the-shelf priors that are themselves obtained without target annotations: a DINOv2 ViT-B/14 backbone [8] pre-trained on a generic image collection, and the CAUSE-TR head [10]. CAUSE [10] is a fully unsupervised semantic-segmentation method that distills DINO patch features into a 90-dimensional concept-clusterbook code by jointly training a Transformer-based projection head with a clusterbook of learnable concept prototypes. The clusterbook organizes the latent space into a small number of class-discriminative regions and is trained on Cityscapes train images using only the self-supervised concept-coherence objective — no human labels at any stage. We use the resulting CAUSE-TR head [10] as a frozen feature provider; together with DINOv2, both priors are obtained without target annotations and no labelled supervision enters either prior. What replaces stereo and motion cues is a Stage-1 generator built from three modules. A semantic branch produces $\hat{S}_x$ by passing x through the frozen DINOv2 backbone followed by the frozen CAUSE-TR head [10], whose semantic code is refined by the Depth-Conditioned Feature Adapter (DCFA) before being clustered into pseudo-classes. An instance branch produces $\hat{I}_x$ by computing Sobel responses on a monocular depth map and extracting connected components inside thing-eligible semantic regions. SIMCF then filters the candidate labels by checking semantic identity, instance geometry, and feature similarity.

3.2 Monocular Pseudo-Label Baseline

Stereo and scene flow give CUPS [17] access to physical-separation cues at pseudo-label time: a 3-D boundary between two adjacent surfaces produces a discontinuity in disparity or in inter-frame motion that separates one instance from another. Monocular depth exposes many of the same physical-separation cues in a single frame, provided that the discontinuity is read at the right spatial scale and only within a region whose semantic identity has already been estimated. Figure 2 shows how the semantic and depth branches provide the raw candidate labels that SIMCF later filters.

The semantic branch is assembled entirely from frozen priors. Each image x is encoded by a frozen DINOv2 ViT-B/14 backbone [8], whose dense patch features are then reduced by a frozen CAUSE-TR head [10] to a 90-dimensional semantic code. We L2-normalize these codes and partition them with k-means on the unlabeled training set into $k = 80$ centroids, assigning each pixel the index of its nearest centroid to obtain $\hat{S}_x$. The cardinality $k = 80$ is deliberately well above the 27-class evaluation taxonomy: collapsing to 27 classes at this stage would erase fine visual modes that the depth-based instance branch later relies on. CUPS reports that this regime monotonically improves panoptic quality as k exceeds the evaluation cardinality [17].

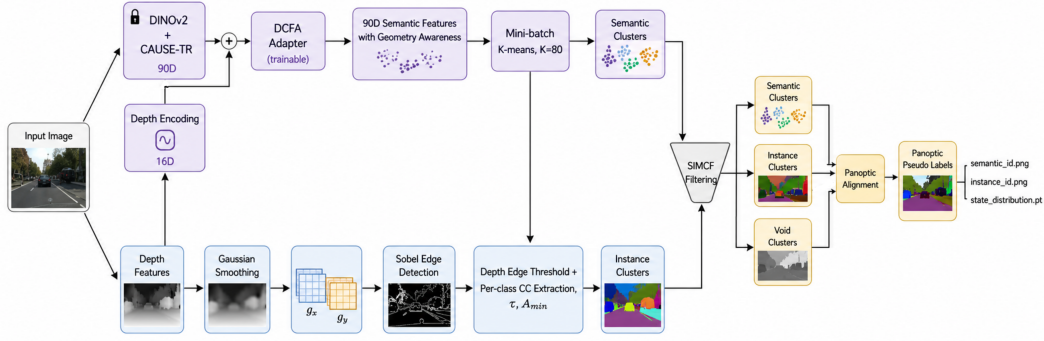


Figure 2: Pseudo-label generation. The semantic branch clusters depth-conditioned CAUSE-TR [10] codes, the instance branch converts monocular depth gradients into connected components inside thing-eligible regions, and SIMCF filters the candidate semantic and instance labels into panoptic pseudo-labels.

With $\hat{S}_x$ in hand, the instance branch reads physical-separation structure from a single monocular depth map $D \in \mathbb{R}^{H \times W}$. We use a frozen DepthPro network [11], chosen because it balances boundary sharpness against surface smoothness. We apply 3×3 Sobel kernels K_x, K_y to D and flag candidate boundaries by thresholding the resulting gradient magnitude,

$$g_D(p) = \sqrt{(K_x * D)^2(p) + (K_y * D)^2(p)}, \quad (2)$$

$$B_D(p) = \mathbf{1}\{g_D(p) > \tau_d\}, \quad \tau_d = 0.20.$$

Pixels with $B_D(p) = 1$ are flagged as candidate boundaries. We set $\tau_d = 0.20$ using label-free pseudo-instance statistics: it keeps the average proposal count and small-component fraction in the range needed by panoptic bootstrapping, whereas sharper thresholds produce many tiny fragments. For example, $\tau_d = 0.01$ produces sharper instance proposals on the depth-only evaluation but over-fragments the training data (~ 57 instances per image, more than half below 1000 pixels). The PQ/SQ/RQ diagnostics in Table 6 are reported post hoc to explain this choice. Connected-component labelling with 8-neighbour adjacency is run only inside thing-eligible semantic regions. Each of the 80 over-clusters is assigned to its nearest concept in the CAUSE [10] 27-class vocabulary by centroid distance in the 90-dimensional code space, and thing-eligibility is read off the CAUSE taxonomy [10] (eight countable categories). The CAUSE 27-class vocabulary [10] itself is produced by the frozen, self-supervised CAUSE-TR head [10] and carries an intrinsic thing/stuff partition that is independent of the Cityscapes evaluation taxonomy: the partition is fixed before any benchmark label is consulted. Both the over-cluster fitting and the CAUSE taxonomy [10] are obtained without target annotations; the Hungarian map between the CAUSE 27-class vocabulary [10] and the Cityscapes 27-class evaluation taxonomy (CUPS protocol [17], see Section 4.1) is computed only at evaluation time and is the only ground-truth-derived component anywhere in the pipeline. Components below $A_{\min} = 1000$ pixels are discarded, and the surviving components are dilated with three iterations of a 3×3 structuring element.

The semantic regions inherited from the frozen CAUSE-TR [10] code are appearance-driven and can split a single physical surface across an illumination change or merge two adjacent same-class objects. Closing this gap requires conditioning the semantic code on the same depth signal that already governs the instance branch.

3.3 Depth-Conditioned Feature Adapter (DCFA)

The semantic regions inherited from the frozen CAUSE-TR [10] code are appearance-driven, and the failure modes are: a single physical surface split across an illumination change, and two adjacent

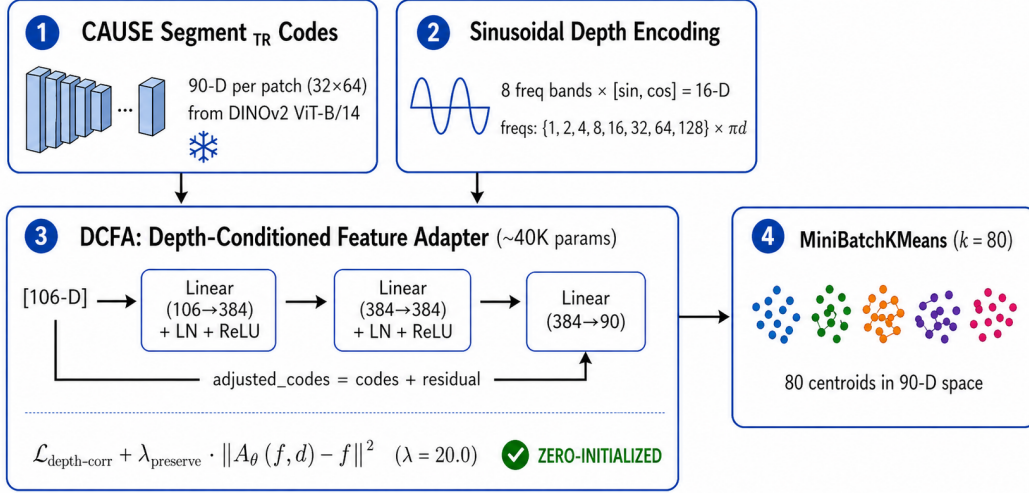


Figure 3: DCFA adapter training. A frozen CAUSE-TR [10] code is adjusted by a depth-conditioned residual before k -means clustering, while a preservation term keeps the adapted code close to the original frozen semantic manifold.

same-class objects merged into one region. Figure 3 summarizes the adapter path. We introduce a small residual adapter that consumes the depth value d_u at pixel u and adds a learned depth-driven shift to the frozen 90-dimensional code z_u , nudging cluster boundaries toward depth-consistent regions where appearance fails.

Conditioning on d_u requires a representation rich enough for a small MLP to read both fine boundary differences and coarse depth ordering. Since $d_u \in [0, 1]$ is a scalar, we lift it through a 16-dimensional sinusoidal positional encoding with eight exponentially-spaced frequencies $\omega_k = 2^k \pi$ for $k = 0, \dots, 7$,

$$e(d_u) = [\sin(\omega_0 d_u), \cos(\omega_0 d_u), \dots, \sin(\omega_7 d_u), \cos(\omega_7 d_u)] \in \mathbb{R}^{16}. \quad (3)$$

The encoding $e(d_u)$ is then passed through a two-layer MLP $r : \mathbb{R}^{16} \rightarrow \mathbb{R}^{90}$ with hidden width $h = 384$, ReLU activation, and a zero-initialized output projection. The MLP produces a depth-only shift that is added residually to the frozen code,

$$\tilde{z}_u = z_u + r(e(d_u)). \quad (4)$$

The construction has approximately 40K trainable parameters, small relative to the ~ 86 M parameters of the frozen ViT backbone. Two design choices keep the adapted code on the frozen CAUSE-TR [10] manifold rather than re-learning a new representation: the output projection is zero-initialized, so that $\tilde{z}_u = z_u$ at the first optimizer step and any departure from the frozen code is acquired gradually through gradient descent; and the preservation term in Eq. 5 with $\lambda_p = 20$ explicitly penalizes drift from z_u , so $\tilde{z}_u$ at convergence is a small, depth-conditioned perturbation of z_u rather than an independently learned code.

Training the adapter calls for a self-supervised signal that pulls the adapted codes of pixels at similar depth toward each other while keeping them on the original CAUSE-TR [10] manifold. DepthG [20] established that depth-feature correlation can guide unsupervised semantic learning when depth governs the loss; we apply the same loss to the adapted code rather than to the backbone weights. For each minibatch image, we sample 1024 pixels and 1024 pixel pairs, using the DepthG [20] depth-similarity kernel with bandwidth $\sigma_d = 0.5$. The adapter minimizes

$$\mathcal{L}_{\text{DCFA}} = \mathcal{L}_{\text{corr}}(\tilde{z}, D) + \lambda_p \mathcal{L}_{\text{preserve}}(\tilde{z}, z), \quad \lambda_p = 20. \quad (5)$$

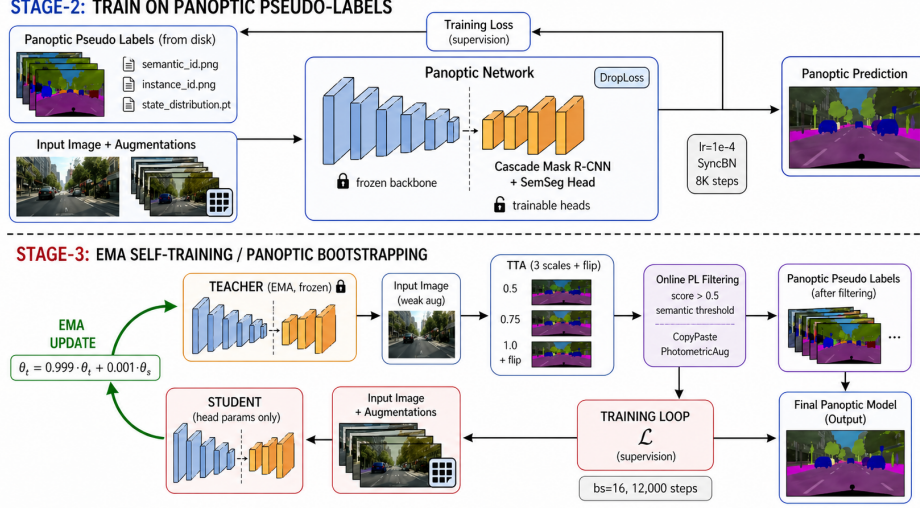


Figure 4: Panoptic bootstrapping and panoptic network training. Filtered pseudo-labels supervise the initial panoptic network, and EMA self-training refines the labels and network across subsequent rounds, following the CUPS [17] training recipe.

Let $\mathcal{U}$ denote the set of $|\mathcal{U}| = 1024$ pixels sampled per image and $\mathcal{P}$ the set of $|\mathcal{P}| = 1024$ pixel pairs sampled from $\mathcal{U}$. Then $\mathcal{L}_{\text{corr}} = \frac{1}{|\mathcal{P}|} \sum_{(i,j) \in \mathcal{P}} \mathcal{K}_{\sigma_d}(D_i, D_j) (1 - \cos(\tilde{z}_i, \tilde{z}_j))$ is the DepthG [20] depth-similarity-weighted correlation loss with bandwidth $\sigma_d = 0.5$, and $\mathcal{L}_{\text{preserve}} = \frac{1}{|\mathcal{U}|} \sum_{u \in \mathcal{U}} \|\tilde{z}_u - z_u\|_2^2$ is the per-pixel mean squared distance between adapted and frozen codes over the same sampled set. Both losses average over the sampled set per image, then average over the minibatch.

3.4 SIMCF: Semantic Instance Mutual Consistency Filtering

DCFA changes the feature space before clustering, but some errors remain after semantic and instance labels are formed. A depth fragment may cut through a single object, a proposal may contain pixels from multiple pseudo-classes, or a semantic assignment may sit at a depth far from the empirical range of its pseudo-class. SIMCF, short for Semantic Instance Mutual Consistency Filtering, is the final algorithm inside G : it takes the raw candidate pair (S_x^0, I_x^0) and returns filtered pseudo-labels $(\hat{S}_x, \hat{I}_x)$ by enforcing consistency between semantic identity, instance geometry, feature similarity, and depth statistics.

Algorithm 1 lists the procedure. The first check prevents mixed-class instance targets, the second repairs depth over-fragmentation using DINOv3 appearance similarity [9], and the third converts depth-improbable pixels into void labels rather than hard negatives. SIMCF is learning-free: it has no trainable parameters, and the four hyperparameters τ_{sim} , η , A_{min} , and the 8-neighbour adjacency are fixed across all experiments and reported in Section 4.1. DCFA remains the only learned component inside the Stage-1 generator, and the filtered labels are passed directly to panoptic bootstrapping.

3.5 Panoptic Bootstrapping and Panoptic Network Training

Filtering completes the Stage-1 generator G . The resulting pseudo-labels $(\hat{S}_x, \hat{I}_x)$ provide the supervision for panoptic bootstrapping, while the RGB image x remains the network input. Figure 4 shows the training path adopted from CUPS. We adopt the panoptic bootstrapping and panoptic network training procedures from CUPS [17], including its loss, copy-paste augmentation, bootstrapping schedule, and EMA self-training. Keeping these procedures fixed across configurations lets the experiments isolate the effect of the pseudo-label source.

Algorithm 1. SIMCF($S_x^0, I_x^0, D, \phi, v; \tau_{sim}, \eta$).

1. $(\hat{S}_x, \hat{I}_x) \leftarrow (S_x^0, I_x^0)$
2. *for each* instance region $R_k = \{p : \hat{I}_x(p) = k\}$ *do*
3. $c_k \leftarrow \arg \max_c |\{p \in R_k : \phi(\hat{S}_x(p)) = c\}|$
4. $s_k \leftarrow \arg \max_s |\{p \in R_k : \hat{S}_x(p) = s, \phi(s) = c_k\}|$
5. *for each* $p \in R_k$ with $\phi(\hat{S}_x(p)) \neq c_k$ *do*
6. $\hat{S}_x(p) \leftarrow s_k$
7. *for each* adjacent proposal pair (a, b) in $\hat{I}_x$ *do*
8. compute mean features $\bar{v}_a, \bar{v}_b$ over R_a, R_b
9. *if* $c_a = c_b$ and $\cos(\bar{v}_a, \bar{v}_b) > \tau_{sim}$ *then*
10. merge R_a and R_b in $\hat{I}_x$
11. *for each* pseudo-class c *do*
12. $P_c \leftarrow \{p : \phi(\hat{S}_x(p)) = c\}$
13. $(\mu_c, \sigma_c) \leftarrow (\text{mean } D(P_c), \text{std } D(P_c))$
14. *for each* pixel p *do*
15. $c \leftarrow \phi(\hat{S}_x(p))$
16. *if* $|D(p) - \mu_c| > \eta \sigma_c$ *then*
17. $\hat{S}_x(p) \leftarrow \text{void}; \quad \hat{I}_x(p) \leftarrow 0$
18. *return* $(\hat{S}_x, \hat{I}_x)$

The panoptic network takes x as input and is trained on $(\hat{S}_x, \hat{I}_x)$. Its cascade, proposal, and panoptic heads use the CUPS-style [17] losses; Section 4.1 specifies the architecture.

Two ingredients of the inherited recipe interact directly with the void labels produced by SIMCF. Following CUPS [17], DropLoss [15] masks losses on void or unreliable regions so that rejected pseudo-labels are not treated as hard negatives, making void a soft abstention rather than a class signal. Self-enhanced copy-paste augmentation operates at the input level: at each training iteration, instances from the pseudo-labels of one image are pasted into another image at random positions. This augmentation increases exposure to thing-class instances.

Once the cascade trainer reaches its bootstrapped checkpoint, three rounds of EMA self-training continue from it. In each round, an exponential-moving-average teacher is maintained over the student weights and used to relabel the unlabeled training set; the student is re-trained on the resulting refined pseudo-labels. The teacher provides a smoother, lower-variance label distribution than G alone supplies, in particular for thing-class boundaries where Stage-1 labels are limited by monocular depth discontinuities. We use the final-round student checkpoint as the network F in Eq. 1.

At test time, F produces a panoptic prediction (S_x, I_x) via the standard panoptic merge of [6] applied to its dense semantic output and cascade instance proposals. Two design choices are non-obvious. First, F predicts directly into the pseudo-class vocabulary on which it was trained, not into the 27-class Cityscapes taxonomy: the Hungarian mapping from pseudo-classes to benchmark classes is computed only at evaluation time and is the only ground-truth-derived component anywhere in the pipeline; neither G nor F ever consults benchmark labels during training. Second, the thing/stuff partition that drives the merge is inherited from the frozen CAUSE-TR [10] taxonomy (§3.2), not from the evaluation taxonomy, so the same partition supervises Stage-1 instance extraction and Stage-3 panoptic merging without taxonomy drift across stages.

4 Experiments

We report experiments on Cityscapes and zero-shot transfer datasets, separating pseudo-label quality from trained-network performance. The evaluation asks whether monocular pseudo-labels can seed the adopted CUPS [17] training procedure, how much DCFA and SIMCF contribute, and where the fixed Cityscapes-derived pseudo-class vocabulary fails.

Table 1: Unsupervised panoptic segmentation on Cityscapes val (all in %, $\uparrow$). “PL inputs” are the inputs available at pseudo-label time. “Eval cls.” is the number of benchmark classes used by the CUPS [17] evaluation protocol; our network predicts $K=80$ pseudo-classes and is aligned to the 27-class taxonomy only at metric time (Section 4.1).

Method	Training data	PL inputs	Eval cls.	PQ	SQ	RQ	PQ Th	PQ St
Supervised [6]	CS	dense labels	–	62.3	81.8	75.1	62.4	62.1
DepthG [20]+CutLER [15]	CS & ImageNet	RGB mono.	27	16.1	45.4	21.1	3.0	25.7
U2Seg [18]	COCO & ImageNet	RGB mono.	800+27	18.4	55.8	22.7	10.2	24.3
S2-UniSeg [19]	SA-1B & CS	RGB mono.	–	25.4	66.7	35.0	–	–
CUPS [17]	CS	RGB+stereo+flow	27	27.80	57.4	35.2	17.7	35.1
Monocular baseline (no DCFA/SIMCF)	CS	RGB mono.	27	32.76	62.57	40.75	–	–
Ours (DCFA+SIMCF)	CS	RGB mono.	27	35.83	62.79	43.78	36.26	35.56
vs. CUPS [17] (prev. scene-centric SOTA)				+8.03	+5.39	+8.58	+18.56	+0.46

4.1 Experimental Setup

Pseudo-label generation and panoptic network training use only Cityscapes train [1] (2,975 images, no annotations); evaluation reports Cityscapes val (500 images) under the 27-class CAUSE [10] pseudo-vocabulary with a Hungarian map to the 27-class Cityscapes label space (NUM_CLASSES=27 in the official CUPS protocol [17], the full Cityscapes label space). The Stage-1 generator G couples a frozen DINOv2 ViT-B/14 backbone [8] and CAUSE-TR head [10] with DCFA ($h=384$, $\lambda_p=20$, $\sim 40K$ trainable parameters), k -means over-clustering at $k=80$, the monocular instance branch ($\tau_d=0.20$, $A_{\min}=1000$), and SIMCF ($\tau_{\text{sim}}=0.85$, $\eta=2.5$); DepthPro [11] supplies the monocular depth map. The panoptic network is a three-stage Cascade Mask R-CNN [14] on a frozen DINOv3 ViT-B/16 backbone [9], trained with CUPS-style [17] panoptic bootstrapping (8K steps, DropLoss [15], copy-paste) followed by three 5K-step EMA self-training rounds. All operating hyperparameters were fixed before consulting Cityscapes annotations; the SIMCF sweep in Table 4 is a retrospective sensitivity analysis, not a model-selection procedure. Full hyperparameter recipes, optimizer schedules, augmentation lists, hardware, and wall-clock budgets are reported in Appendix ?? of the supplementary material.

4.2 Main Cityscapes Results

Table 1 records two comparisons on Cityscapes val: published external references, including CUPS [17], and an in-paper controlled comparison that holds the Stage-2/3 trainer fixed. Our method reaches 35.83 PQ, a +3.07 PQ gain over the matched monocular baseline and a +8.03 PQ gain over the published stereo+flow CUPS reference [17] under the reported CUPS evaluation protocol [17] (+5.39 SQ, +8.58 RQ, +18.56 PQTh, +0.46 PQSt). The supervised row (62.3 PQ) is an oracle upper bound.

The two “ours” rows isolate the contribution of DCFA+SIMCF: both share the same Stage-2/3 recipe, identical panoptic bootstrapping and EMA self-training, and identical Cascade Mask R-CNN backbone; only the Stage-1 pseudo-label source differs. The monocular baseline is trained on raw $K=80$ pseudo-labels (24.54 pseudo-PQ), without DCFA or SIMCF, with Stage-1 inputs differing only in (a) raw 90-D CAUSE-TR [10] semantic code (no DCFA residual), and (b) DepthPro connected components at $\tau_d = 0.20$ with no SIMCF filtering. DCFA+SIMCF add +3.07 PQ to the trained-network result in this controlled comparison.

The final model after Stage-2 panoptic bootstrapping and three rounds of Stage-3 EMA self-training reaches 35.83 PQ, 62.79 SQ, and 43.78 RQ on Cityscapes val. The improvement over the pseudo-label generator is concentrated in recognition rather than mask quality, indicating that the panoptic bootstrapping and EMA self-training stages principally amplify class recall while preserving the mask geometry already produced by the Stage-1 generator. The trained-model mIoU of 44.56% is reported under the 19-class semantic-segmentation evaluation taxonomy used historically for unsupervised mIoU comparison.

The aggregate gain is not uniform: large stuff regions and large vehicles combine high SQ with high RQ, while thin structures and rare classes are recognition-limited. The full per-class breakdown is reported in Appendix ?? of the supplementary material and summarized alongside the failure analysis in Section 4.4.

Table 2: Pipeline progression on Cityscapes val. Each row adds one component to the row above.

#	Pipeline stage	PQ
1	Stage-1 pseudo-labels (raw, no DCFA, no SIMCF)	24.54
2	+ DCFA + SIMCF (our Stage-1 contributions)	25.85
3	+ Stage-2 panoptic bootstrapping	28.40
4	+ Stage-3 EMA self-training (final)	35.83

Table 3: Pseudo-label component complementarity at fixed $\tau_d = 0.20$ DepthPro instances. “Raw $K=80$ ” is the no-DCFA, no-SIMCF baseline. mIoU is the Cityscapes val 19-class semantic-segmentation mIoU; SIMCF filters instance proposals and does not modify the semantic prediction, so the SIMCF-only and DCFA+SIMCF rows inherit the mIoU of the underlying semantic code.

Variant	DCFA	SIMCF	PQ	SQ	RQ	mIoU
Raw $K=80$	–	–	24.54	60.0	33.6	52.69
DCFA only	✓	–	25.22	61.0	34.0	55.29
SIMCF only	–	✓	25.27	57.5	34.4	52.69
Both	✓	✓	25.85	58.3	34.9	55.29

4.3 Ablations

Pipeline progression. Table 2 traces how PQ accumulates across the four pipeline stages. Stage-1 raw pseudo-labels reach 24.54 PQ; adding our two Stage-1 contributions (DCFA + SIMCF) lifts pseudo-label PQ to 25.85 (+1.31). Stage-2 panoptic bootstrapping then trains a Cascade Mask R-CNN on these pseudo-labels and reaches 28.40 PQ, and Stage-3 EMA self-training carries the network to its final **35.83 PQ** on Cityscapes val. Two observations follow. First, every stage of the pipeline contributes a meaningful uplift; the largest single jump is the trained-network step, which converts pseudo-label structure into recognition recall. Second, the contribution of DCFA+SIMCF compounds through training: the +1.31 PQ Stage-1 advantage grows to +3.07 PQ at the trained-network level, indicating that better pseudo-labels are not only better at pseudo-label time but also amplify under the Stage-2/3 procedure.

Component contribution. Table 3 reports DCFA $\times$ SIMCF at the fixed training-data instance configuration ($\tau_d = 0.20$). All four cells share the same depth instances; only the pre-clustering adapter and the post-instance filter are toggled. The load-bearing observation is the +1.31 PQ jump from Raw $K=80$ (24.54) to Both (25.85), and the +0.58/+0.63 PQ separation between Both and either single component, which is consistent with the two corrections targeting different failure modes (DCFA reshapes the feature space, SIMCF filters the resulting labels) and combining super-additively. The single-component rows DCFA-only (25.22) and SIMCF-only (25.27) are within 0.05 PQ of each other and should not be read as a strict ordering between the two mechanisms; the relevant claim is complementarity, evidenced by the move from either single component to Both.

SIMCF threshold sensitivity. SIMCF exposes two thresholds: τ_{sim} for adjacent-proposal feature merging (Step B) and η for depth-implausibility void rejection (Step C). To verify that SIMCF’s contribution is not the product of threshold tuning, we sweep both $\pm \sim 12\%$ around the paper setting and re-evaluate pseudo-label quality on the full Cityscapes train set ($N = 2975$). Table 4 reports the result. Pseudo-label PQ varies within a 0.71-point band (25.27–25.98) across the sweep, smaller than the +1.31 PQ super-additive gap from either single component to Both reported in Table 3. Tighter thresholds slightly improve PQ (more aggressive merging recovers more depth over-fragmentations); looser thresholds degrade modestly. The mechanism is therefore robust to moderate threshold perturbation rather than tuned-on-test.

DCFA conditioning mechanism. The novelty claim in Section 2 contrasts DCFA’s architectural depth conditioning with DepthG-style [20] loss-side conditioning. We separate the contribution of the two axes empirically (Table 5). *DepthG* [20] (*drop-in*) loads the official DepthG checkpoint [20] (DINO ViT-B/8 + STEGO-style cluster head trained with depth-correlation loss on Cityscapes train, $\text{dim}=100$, ZoeDepth) and uses it as a Stage-1 generator under our identical downstream pipeline (k-means $k=80$, DepthPro depth-CC instances, eval). *Loss-side* $P(z)$ replaces DCFA’s

Table 4: SIMCF threshold sensitivity at fixed depth instances ($\tau_d = 0.20$, DepthPro). Pseudo-label PQ is computed on the full Cityscapes train set under the CUPS [17] 27-class protocol; discussed in Section 4.3.

Setting	τ_{sim}	η	PQ	SQ	RQ	mIoU
tighter	0.75	2.0	25.98	58.79	34.55	56.08
baseline (paper)	0.85	2.5	25.87	58.38	34.83	56.17
looser	0.95	3.0	25.27	57.63	34.10	56.22

Table 5: Conditioning-mechanism decomposition at the Stage-1 pseudo-label level (Section 4.3). All rows share the same downstream pipeline; only how depth enters the model varies.

Method	Conditioning	PQ	SQ	RQ	mIoU
Raw $K=80$ (paper substrate)	none	24.54	60.00	33.60	52.69
DepthG [20] (official, ViT-B/8)	loss, end-to-end	13.33	33.39	18.56	–
Loss-side $P(z)$ on CAUSE [10]	loss, no preserve	12.62	55.35	18.95	–
Loss-side $P(z)$ + preserve	loss + preserve	23.83	56.54	32.76	–
DCFA (ours)	residual + preserve	25.22	61.00	34.00	55.29

depth-conditioned residual with a learnable projection $P : \mathbb{R}^{90} \rightarrow \mathbb{R}^{90}$ ($P(z) = z + \text{MLP}(z)$, 70K parameters, no depth as architectural input) trained on cached CAUSE [10] codes with only the depth-correlation loss $\mathcal{L}_{\text{corr}}(P(z), D)$. We further isolate the preservation axis with *Loss-side* $P(z)$ + *preserve*, which adds $\lambda_p \mathcal{L}_{\text{preserve}}$ ($\lambda_p = 20$) to the loss-side objective.

The decomposition reads as follows. Without preservation, both loss-side methods collapse far below the raw baseline (DepthG [20] drop-in -11.21 PQ; loss-side without preserve -11.92 PQ), confirming that any unconstrained re-shaping of a frozen unsupervised concept-clusterbook code degrades the embedding structure that downstream k -means relies on. Adding preservation rescues most of the structure (loss-side with preserve reaches 23.83 PQ), but is still -0.71 PQ below the raw baseline and -1.39 PQ below DCFA. The remaining $+1.39$ PQ gap between loss-side+preserve and DCFA, at matched parameter budget and matched preservation strength, is the empirical contribution of *architectural* depth conditioning – depth as an input to the conditioning network, not just to the loss. The two axes – architectural input and explicit manifold preservation – are both required for the $+0.68$ PQ over raw, and neither is a strict substitute for the other.

Depth model. Table 6 reports representative depth-source and threshold runs for which the evaluations provide aggregate SQ and RQ. The rows are not used as a single leaderboard because some runs use the raw $K=80$ semantic map and others use the DCFA semantic map; they are used to separate mask quality from recognition quality while choosing the training-data configuration.

The table shows why PQ alone is not a sufficient diagnostic for the depth branch. The sharper $\tau = 0.01$ DepthPro setting has stronger standalone PQ/SQ/RQ, but it produces a fragmented training set with roughly 57 instances per image and more than half below 1000 pixels. The $\tau = 0.20$ setting has lower standalone PQ but produces larger training targets, roughly 22 instances per image, which is the configuration used for panoptic bootstrapping. DA v3 at the same training-data threshold reaches a similar RQ, indicating that the choice is governed by the training signal produced by the depth branch rather than by the stuff/thing aggregate split.

Table 6: Depth and threshold diagnostics with aggregate SQ/RQ.

Depth	Sem.	τ	PQ	SQ	RQ
SPIdeth	raw $K=80$	0.20	26.74	71.88	31.41
DA v3 [13]	raw $K=80$	0.03	27.37	73.44	35.66
DA v3 [13]	DCFA	0.20	26.44	61.37	35.83
DepthPro [11]	DCFA	0.01	27.81	63.07	37.40
DepthPro [11]	DCFA	0.20	26.13	60.63	35.80

Table 7: Cross-dataset generalization on driving-scene benchmarks (PQ/SQ/RQ in %, $\uparrow$). Published baselines from [17] Tab. 2. “–” marks datasets a baseline does not report. [§]KITTI 19-class and 27-class are identical (KITTI GT only contains the 19 standard classes); [¶]Waymo via CUPS [17] Waymo loader; [†]Mapillary via paper-defined remap. Protocol details discussed in Section 4.4; OOD sanity checks (MOTS, COCO-Stuff-27) in Appendix ??.

Method	KITTI [3]			Waymo V2 [4] [¶]			Mapillary v2 [†]		
	PQ	SQ	RQ	PQ	SQ	RQ	PQ	SQ	RQ
Supervised [6]	31.9	71.7	40.4	31.5	70.1	40.9	–	–	–
DepthG [20]+CutLER [15]	11.0	34.5	13.8	13.4	37.3	17.0	–	–	–
U2Seg [18]	20.6	52.9	25.2	19.8	50.8	23.4	–	–	–
CUPS [17]	25.5	58.1	32.5	26.4	60.3	33.0	–	–	–
Ours (DCFA+SIMCF), 19-class[§]	34.85	62.43	42.91	43.51	76.82	50.54	39.19	74.59	47.74
Ours (DCFA+SIMCF), 27-class[§]	34.85	62.43	42.91	–	–	–	27.79	59.64	35.43
<i>vs. CUPS [17]</i>	<i>+9.4</i>	<i>+4.3</i>	<i>+10.4</i>	<i>+17.1</i>	<i>+16.5</i>	<i>+17.5</i>	–	–	–

4.4 Generalization and Failure Analysis

Our trained network transfers across the driving class space without retraining. KITTI reaches 34.85 PQ, surpassing CUPS [17] by +9.35 PQ; Waymo V2 reaches 43.51 PQ, surpassing CUPS [17] by +17.11 PQ; Mapillary v2 reaches 39.19 PQ. We report each dataset under the 19-class CUPS cross-dataset protocol [17] used by CUPS for KITTI, and report KITTI and Mapillary again under the 27-class CUPS in-domain protocol [17] for paper-internal consistency (Table 7). KITTI’s panoptic ground truth contains only the 19 standard Cityscapes classes, so 19-class and 27-class evaluations yield identical numbers (34.85 PQ in either taxonomy). Mapillary contains ground truth for all eight Cityscapes-27 extras and is genuinely protocol-sensitive: 27-class PQ is 27.79 because eight low-coverage extras (parking, rail track, guard rail, bridge, tunnel, polegroup, caravan, trailer; 4.42% of GT pixels combined) enter the average rather than mapping to void. Waymo V2 is evaluated through the CUPS Waymo pathway [17], which maps raw Waymo panoptic labels to Cityscapes-19 over the 16 supported classes. Out-of-distribution sanity checks on a thing-only label space (MOTS [2]) and a vocabulary-mismatched object-centric dataset (COCO-Stuff-27 [5]) are deferred to Appendix ?? of the supplementary material; the COCO-Stuff-27 result (7.83 PQ) is dominated by categories absent from a Cityscapes-derived pseudo-label space and indicates the vocabulary boundary of in-domain training rather than a transfer failure of the architecture.

Per-class failure analysis. The aggregate 35.83 PQ is supported by twenty active classes; large stuff regions (road, sky, vegetation, building, all > 83 PQ) and large vehicles (car, bus, train, all > 70 PQ) dominate, while six vocabulary-dead classes (parking, guard rail, tunnel, polegroup, caravan, trailer) lie outside the standard Cityscapes-19 panoptic benchmark and receive too few positive examples in the $K=80$ over-cluster vocabulary. The full per-class PQ/SQ/RQ breakdown for all 27 classes is reported in Appendix ?? of the supplementary material. These vocabulary-coverage cases reflect a property of the pseudo-label vocabulary rather than the network architecture, and their effect on the aggregate is bounded.

Two qualitative metrics from the feature-guided merge step of SIMCF close the analysis. Average pseudo-instance count per image drops from 44 before the merge to 22 after, median instance size grows from 5,502 to 14,965 pixels, and stuff contamination falls from 50.7% to 28.0%. Many depth gradients are intra-object surface changes rather than object boundaries, and merging fragments whose semantic identity and dense appearance agree recovers the single-object structure that monocular depth had over-fragmented.

5 Discussion

The results support three observations. First, cross-modal agreement improves monocular pseudo-labels at both feature and label levels. DCFA changes the semantic code before clustering, while SIMCF filters candidate labels after instance extraction. The component-complementarity table

in Section 4.3 shows that the combined system improves pseudo-label PQ by +1.31 over the raw monocular baseline, more than either component alone. Under the fixed panoptic bootstrapping and network-training procedure, this pseudo-label improvement corresponds to a +3.07 PQ gain over the corresponding monocular baseline.

Second, monocular pseudo-labels are competitive in the driving-scene setting studied here. CUPS [17] constructs pseudo-labels from stereo depth and unsupervised scene flow; our generator uses a single monocular depth map and frozen self-supervised features. The experiments show that this replacement is sufficient to train a strong panoptic network under the adopted CUPS procedure [17]. The conclusion is scoped to Cityscapes-style driving scenes, where DepthPro [11] depth and frozen DINO-family features [8, 9] supply much of the structure that stereo and motion provided in CUPS [17], and the controlled comparison in Table 1 between the monocular baseline and our DCFA+SIMCF row attributes a +3.07 PQ delta to the proposed pseudo-label generator under a fixed Stage-2/3 procedure.

Third, the per-class table cleanly separates mask quality from recognition: person reaches SQ=71.4 and rider SQ=62.6, both above the 60-point band that classifies the per-class match as geometrically correct, while their RQ values (18.7 and 36.6 respectively) reflect the recognition bottleneck for closely-spaced object classes. The split confirms that DCFA+SIMCF act on the parts of the pipeline they are designed to address — the frozen semantic code and candidate-label consistency — and that further gains on closely-spaced pedestrians lie upstream in the depth and semantic priors.

6 Limitations

Dead classes remain the main in-domain limitation: six classes in the 27-class pseudo-vocabulary receive too few reliable pseudo-labels, leaving them effectively absent after Hungarian alignment (per-class numbers in Appendix ??). This vocabulary limitation also explains the main out-of-domain failure. The Cityscapes-derived pseudo-label space transfers within driving scenes, but it fails on object-centric COCO-Stuff-27, where many categories are outside the supervision represented by the learned vocabulary; out-of-distribution sanity checks on COCO-Stuff-27 (vocabulary mismatch) and MOTS (thing-only label space) are reported in Appendix ??.

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